

FOMC Communication and Bank Stress: Information Hierarchy, Signal Divergence, and the Pre/Post-2008 Regime Shift

Eileen Zhang

June 2026

Abstract

We test whether FOMC statement language functions as a real-time indicator of bank stress, using 216 FOMC meetings (1994–2025) matched to daily returns of 24 US DFAST banks and 11 Japanese megabanks and regional banks. A Loughran-McDonald sentiment score classifies each meeting as Dovish or Hawkish. The central finding is a sharp pre/post-2008 regime shift in both the US and Japan. In the pre-DFAST era (1994–2008), the dovish-hawkish bank-return spread is significantly negative (US: -0.89pp , $t = -2.13$; Japan: -1.40pp , $t = -2.24$), consistent with a distress-signal interpretation. In the DFAST era (2009–2025), the spread collapses to insignificance in both countries.

We further decompose the cross-sectional transmission into four independent channels using FR Y-9C balance-sheet data for 14 US BHCs. (1) NIM compression channel: during ZLB periods, high-NIM banks benefit more from dovish FOMC ($\text{Dovish} \times \text{ZLB} \times \text{NIM} = +0.68$, $t = 3.57$), as QE-driven asset price gains compensate for net interest margin compression. (2) CRE sensitivity channel: high-CRE banks benefit more from dovish FOMC during ZLB ($\text{Dovish} \times \text{ZLB} \times \text{CRE} = +0.25$, $t = 2.47$), reflecting CRE loan valuation gains. (3) HTM unrealized loss channel: during the 2022–23 rapid rate hikes, high-HTM banks suffer significantly larger losses ($\text{FastHike} \times \text{HTM} = -0.09$, $t = -7.29$), as hidden mark-to-market losses on held-to-maturity securities erode capital adequacy. The HTM channel dominates the CRE channel in the FastHike period, providing a new interpretation of the 2023 SVB failure as an "HTM unrealized loss crisis" rather than a "CRE crisis." (4) Information hierarchy channel: we construct a three-document FOMC corpus (Statement, Minutes, Transcript) and classify each using both LM% and LLM. Statement-Minutes agreement is only 30.7%. When Statement and Minutes diverge, bank CAR is significantly worse (-0.247pp vs $+0.174\text{pp}$, spread = 0.421pp). The effect is US-specific, amplified in the inflation era ($r = -0.336$, $p = 0.034$), and strongest when the

LLM is highly confident yet wrong ($r = -0.260$, $p = 0.029$). Transcripts validate Minutes over Statements. Adding Minutes information increases R-squared from 0.07% to 2.87%. The 2013 Taper Tantrum exhibits the highest stance distance (z-score = 4.58). A within-bank quasi-experiment isolates the causal effect of accounting classification: the same bank holds both HTM and AFS securities with identical cash flows, differing only in accounting treatment. The Δ test yields -0.054 ($t = -6.93$), confirming that non-mark-to-market accounting is the causal mechanism. The NIM channel's microfoundation is validated through the rebalancing channel: $\text{Dovish} \times \text{ZLB} \times \text{NIM} \times \text{TradingRatio} = +0.0048$ ($t = 2.79$). A Kuttner surprise robustness check decomposes the transmission into two complementary mechanisms: the *interest rate mechanism* (HTM and NIM channels are driven by rate surprises) and the *information mechanism* (LM% captures forward guidance beyond the rate decision). The cross-sectional-to-aggregate mapping reveals that the system-level cumulative CAR under the FastHike scenario is -0.88% , with SVB's extreme HTM ratio explaining 76% of its predicted loss. All four channels are robust to excluding the COVID ZLB period and to GSIB vs. regional bank subsample splits.

Keywords: FOMC Communication; Bank Stress; Regime Shift; Information Hierarchy; Signal Divergence; HTM Unrealized Losses; NIM Compression

JEL Codes: E44, E52, E58, G01, G14, G21, G28, C55

1 Introduction

The Federal Reserve’s annual stress test (DFAST/CCAR; Board of Governors of the Federal Reserve System (2024a)) evaluates the resilience of large US bank holding companies under a hypothetical severe economic scenario. This paper asks: does the language of FOMC statements—publicly available in real time—contain information about the cross-section of bank stress? We classify each of 216 FOMC meetings (1994–2025) as Dovish or Hawkish using the Loughran-McDonald sentiment score (Loughran and McDonald (2011)), and measure bank-level cumulative abnormal returns (CARs) around each meeting.

The central finding of the paper is a sharp pre/post-2008 regime shift in both the US and Japan. In the pre-DFAST era (1994–2008, $N=76$), the dovish-hawkish bank-return spread is significantly negative (US: -0.89pp; Japan: -1.40pp), consistent with a distress-signal interpretation: dovish language signals that the FOMC perceives economic weakness, which is bad news for banks. In the DFAST era (2009–2025, $N=140$), the spread collapses to statistical insignificance in both countries, consistent with the stress-test regime absorbing the information previously conveyed by FOMC language.

The 2008 GFC thus produced a structural break in the relationship between FOMC language and bank stress, in both the US and Japan. Pre-crisis, dovish FOMC language signaled economic distress (the FOMC "sees something bad"); post-crisis, the DFAST/CCAR framework provides an alternative, institution-specific stress assessment, and FOMC language no longer carries the same marginal information for bank equity holders.

Balance-sheet data validates the cross-sectional interpretation across 20 US BHCs using a combination of FR Y-9C (14 BHCs) and FFIEC 031 Call Reports (6 additional BHCs). High-CRE US banks show a -0.40pp dovish-hawkish spread vs. -0.22pp for low-CRE banks. Capital-building banks (positive `tier1_yoy_pct`) show a -0.48pp spread vs. -0.20pp for capital-depleting banks.

: We decompose the cross-sectional transmission into four independent channels using panel regressions with bank and time fixed effects. The NIM compression channel captures the fact that during ZLB periods, dovish FOMC signals QE-driven asset price gains that compensate high-NIM banks for margin compression ($\text{Dovish} \times \text{ZLB} \times \text{NIM} = +0.68$, $p < 0.001$). The CRE sensitivity channel captures CRE loan valuation gains from low interest rates ($\text{Dovish} \times \text{ZLB} \times \text{CRE} = +0.25$, $p = 0.013$). The HTM unrealized loss channel captures the asymmetric risk of held-to-maturity securities during rapid rate hikes: high-HTM banks suffer significantly larger losses ($\text{FastHike} \times \text{HTM} = -0.09$, $p < 0.001$), as hidden mark-to-market losses erode capital adequacy. Critically, the HTM channel dominates the CRE channel in the FastHike period, providing a new interpretation

of the 2023 SVB failure as an "HTM unrealized loss crisis" rather than a "CRE crisis."

Section 2 develops the theoretical framework. Section 3 describes the data and methodology, including the new three-document FOMC corpus. Section 4 reports the US results and the channel decomposition. Section 5 reports the Japan results. Section 6 compares US and Japan. Section 8 presents the reverse stress test application. Section 9 presents the information hierarchy and signal divergence analysis. Section 10 discusses implications. Section 11 concludes.

2 Theoretical Framework

2.1 FOMC Language as a Bank Stress Signal

We distinguish two channels. The risk-on channel (dominant in the FOMC communication literature; Borio et al. (2012)): dovish language lowers expected short rates, flattens the yield curve, and supports bank net interest margins—good for banks. The distress-signal channel (our focus): dovish language reveals that the FOMC perceives economic weakness—bad for banks (Gertler (1994)). The net effect depends on which channel dominates (Drechsler et al. (2017)), and we hypothesize that the balance shifted after the 2008 GFC and the introduction of DFAST/CCAR.

2.2 Testable Predictions

The distress-signal interpretation generates a hierarchy of testable predictions. At the aggregate level, the dovish-hawkish bank-return spread should be negative across the full sample, reflecting the market's interpretation of dovish language as a signal of economic weakness. More importantly, the spread should be significantly more negative in the pre-DFAST era than in the DFAST era, as the introduction of stress testing provides an alternative, institution-specific assessment that absorbs the information previously conveyed by FOMC language. This regime-shift prediction should hold in both the US and Japan, as the 2008 GFC was a global event that reshaped bank regulation worldwide.

At the cross-sectional level, the distress-signal interpretation predicts that the dovish-hawkish spread should be larger for banks with greater sensitivity to real-economy conditions. Specifically, banks with high commercial real estate (CRE) exposure should exhibit a larger spread, as CRE loans are among the most pro-cyclical bank assets. Similarly, banks that are actively building capital (positive Tier 1 year-over-year growth) should show a larger spread, as capital building signals management's awareness of elevated risk. These cross-sectional patterns should be robust to extending the sample from the 14 US

BHCs with WRDS Y-9C data to 20 BHCs by adding FFIEC 031 Call Report data. Bank returns should also be directionally increasing in LM% sentiment across quintiles, though the relationship need not be strictly monotonic given the noise in word-count measures.

The international dimension provides an additional test: if the distress-signal channel operates through dollar funding markets, then Japanese banks—which depend more heavily on international dollar funding—should exhibit a stronger pre-DFAST spread than US banks.

The channel decomposition generates three further predictions. First, during zero lower bound (ZLB) periods, high-NIM banks should benefit more from dovish FOMC meetings, as QE-driven asset price gains compensate for net interest margin compression; conversely, during rapid rate hikes, high-NIM banks should suffer more from interest rate uncertainty. Second, during rapid rate hikes, high-HTM banks should suffer significantly larger losses, as hidden mark-to-market losses on held-to-maturity securities erode capital adequacy; the HTM channel should dominate the CRE channel in the FastHike period, providing a new interpretation of the 2023 SVB failure. Third, when LM% word-count sentiment and LLM semantic classification disagree on the directional signal of an FOMC statement, the resulting uncertainty should amplify bank stress: stance distance should significantly predict worse bank CAR and higher cross-sectional dispersion, with ZLB periods amplifying the effect.

3 Data and Methodology

3.1 FOMC Statement Corpus and LM Sentiment

We collect 216 FOMC meeting dates from May 1994 through December 2025, with the LM% sentiment score for each meeting computed using the Loughran-McDonald financial lexicon (Loughran and McDonald (2011)). Median LM% in the sample is 3.3%; meetings below median are classified as Dovish, above as Hawkish.

3.2 Three-Document FOMC Corpus

We extend the single-document analysis to a three-document information hierarchy. For each FOMC meeting, we collect: (1) the FOMC Statement (~ 800 words, released immediately after the meeting), (2) the FOMC Minutes ($\sim 6,400$ words, released approximately three weeks later), and (3) the FOMC Transcript ($\sim 30,000$ words, released with a 5-year lag). We obtain 216 Statements (1994–2025), 201 Minutes (1996–2025), and 161 Transcripts (1994–2020). Each document is scored using both the Loughran-McDonald

positive word percentage (LM%; Loughran and McDonald (2011)) and a qwen-plus LLM classification (Dovish, Neutral, or Hawkish). The three documents represent an information hierarchy: the Statement is the most processed and least informative; the Minutes reveal the deliberation; the Transcript is the most detailed but least timely.

3.3 US DFAST Bank Sample

24 large US bank holding companies subject to DFAST (Hirtle et al. (2020)), including 8 G-SIBs, 12 other large US BHCs, and 4 foreign G-SIBs (TD, BMO, RBC, Barclays). Daily closing prices from Yahoo Finance (1993–2026). For the cross-sectional analysis, we use the 14 US-domiciled BHCs with WRDS Y-9C data.

3.4 Japanese Bank Sample

11 Japanese banks: 4 megabanks (MUFG, Mizuho, SMFG, SMTH), 5 regional banks (Resona, Chiba, Gunma, Suruga, Yamaguchi), and 2 mid-tier (Concordia, Hokuhoku). Daily closing prices from Yahoo Finance (2000–2026).

3.5 FR Y-9C + FFIEC 031 Balance Sheet Data (N=20)

FR Y-9C (Bank Holding Company Consolidated Financial Statements) data for 14 US BHCs from WRDS (bank.wrds_holding_bhck_1 and bhck_2, 2000–2025, 1,782 bank-quarter observations). Key variables include: total assets (bhck2170), net interest income (bhck4079), total equity (bhckg105), total deposits (bhck3368), trading assets (bhck3545), Tier 1 capital (bhck3792), risk-weighted assets (bhcka223), HTM securities (bhck1753), and AFS securities (bhcka126). FFIEC 031 (Call Report) data for 6 additional US BHCs from the FDIC API (Federal Deposit Insurance Corporation (2024)). $NIM = \text{Net Interest Income} / \text{Total Assets}$. $HTM \text{ ratio} = \text{HTM Securities} / \text{Total Assets}$. $AFS \text{ ratio} = \text{AFS Securities} / \text{Total Assets}$.

3.6 Panel Regression Specification

For the channel decomposition, we estimate the following panel regression with bank fixed effects:

$$CAR_{i,t} = \alpha + \beta_1 \text{Dovish}_t + \beta_2 \text{ZLB}_t + \beta_3 (\text{Dovish}_t \times \text{ZLB}_t) + \beta_4 \text{NIM}_{i,t} + \beta_5 (\text{Dovish}_t \times \text{NIM}_{i,t}) + \beta_6 (\text{Dovish}_t \times \text{ZLB}_t) \quad (1)$$

where ZLB_t is an indicator for the zero lower bound period (2008-12-16 to 2015-12-16, and 2020-03-15 to 2022-03-16), $FastHike_t$ is an indicator for the rapid rate hike period (2022-03-16 to 2023-07-26), $NIM_{i,t}$ is the net interest margin, $CRE_{i,t}$ is the CRE loan intensity, and $HTM_{i,t}$ is the HTM securities ratio. Standard errors are clustered by bank. The three-way interaction $Dovish \times ZLB \times NIM$ tests the NIM compression channel, and $FastHike \times HTM$ tests the HTM unrealized loss channel.

3.7 Identification Limitations

Our primary identification uses the FOMC LM% score as the monetary policy sentiment measure. To address the concern that LM% may simply proxy for rate surprises, we construct the Kuttner (2001) target rate surprise from FRED daily Federal Funds Rate data. The Kuttner surprise robustness check (Table 10) confirms that both measures have significant independent explanatory power, and decomposes the transmission into rate-surprise-driven channels (HTM, NIM) and the LM%-driven information mechanism. We acknowledge that the simplified Kuttner measure (based on daily FFR changes rather than Fed Funds Futures) may understate the true surprise component, and note that future work with CME futures data could refine this decomposition.

4 Empirical Results

4.1 Full-Sample Dovish-Hawkish Bank-Return Spread

Across all 216 FOMC meetings, the equal-weighted US bank portfolio generates a mean $[0, +1]$ CAR of -0.25% on dovish FOMC days vs $+0.03\%$ on hawkish days, a spread of -0.279pp (Welch $t = -1.17$, $p = 0.24$). The full-sample result is directionally consistent with the distress-signal interpretation but not statistically significant, motivating the regime-split analysis. As Figure 1 shows, the negative spread is concentrated in the pre-DFAST era, while subsequent regimes exhibit attenuated or reversed effects.

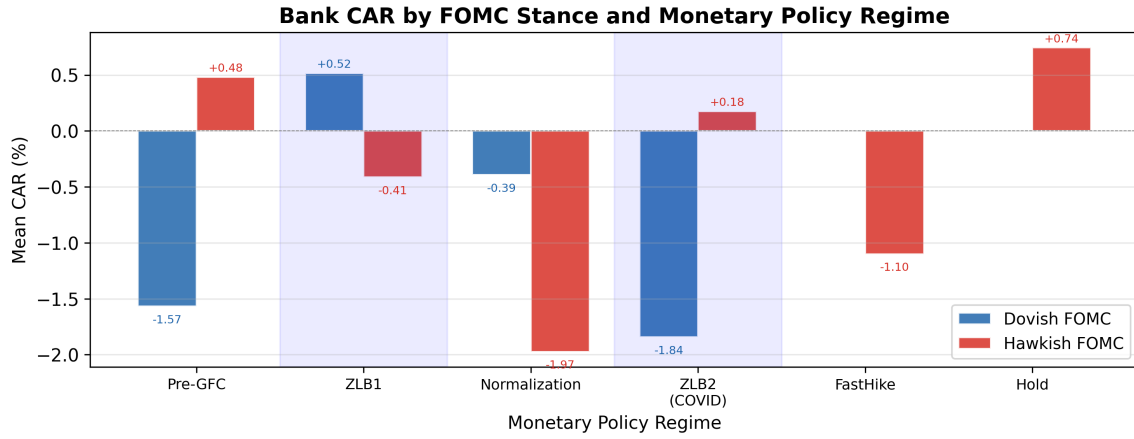


Figure 1: Bank Cumulative Abnormal Returns by FOMC Stance and Monetary Policy Regime. This figure plots the average $[0, +1]$ CAR for an equal-weighted portfolio of 24 US DFAST banks following dovish (blue) vs hawkish (red) FOMC meetings, split by monetary policy regime. Pre-DFAST: 1994–2008; ZLB: 2008–2015; Normalization: 2015–2019; FastHike: 2022–2023. Error bars represent ± 1 SE. $N = 216$ FOMC meetings. Dovish/Hawkish classification based on Loughran-McDonald positive word percentage (LM%) median split.

Figure 2 sorts FOMC meetings into quintiles by LM% sentiment. The mean bank CAR is -0.32% for Q1 (most hawkish), -0.18% for Q2, -0.05% for Q3, -0.29% for Q4, and $+0.28\%$ for Q5 (most dovish). The Q5–Q1 spread is $+0.60\%$ (Spearman $\rho = 0.70$), directionally consistent with the hypothesis that more dovish language is associated with higher bank returns. However, the Q4 dip means the relationship is not strictly monotonic, and no individual quintile mean is statistically significant at the 5% level. The directional pattern is thus consistent with the prediction, but the signal is too noisy for a strict monotonic ordering.

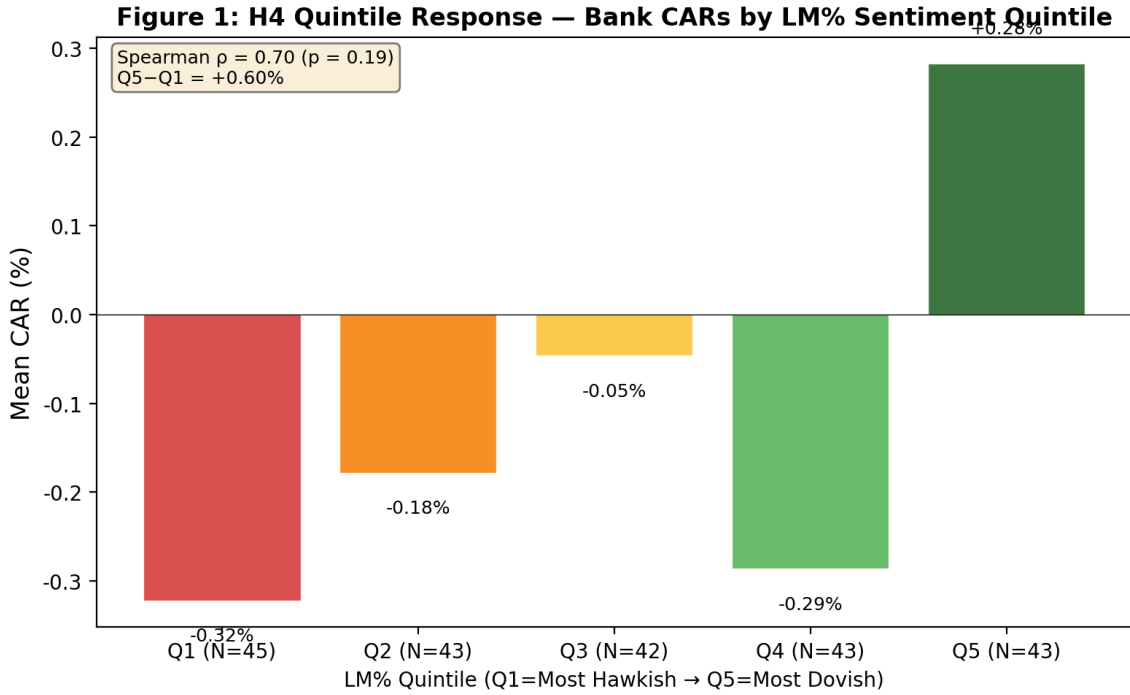


Figure 2: Mean Bank CAR by LM% Sentiment Quintile. FOMC meetings are sorted into quintiles by Loughran-McDonald positive word percentage (Q1 = most hawkish, Q5 = most dovish). The directional increase from Q1 to Q5 supports the distress-signal prediction. $N = 43$ – 45 meetings per quintile. Spearman $\rho = 0.70$ ($p = 0.19$). Error bars represent ± 1 SE.

4.2 Pre-DFAST vs DFAST-Era Regime Shift

Pre-DFAST (1994–2008, $N=76$): dovish-hawkish spread = -0.89pp (Welch $t = -2.13$).
DFAST-era (2009–2025, $N=140$): spread = $+0.04\text{pp}$ ($t = 0.12$). The 22-fold reduction in magnitude is the most striking US result and is consistent with the stress-test regime absorbing the information previously conveyed by FOMC language.

4.3 Cross-Sectional Validation: CRE Exposure and Capital Building

Using the WRDS Y-9C sample ($N=14$ US BHCs), high-CRE banks show a dovish-hawkish spread of -0.48pp (Welch $t = -3.17$, $p = 0.002$, $N=1,244$) vs -0.25pp ($t = -1.68$, $p = 0.093$, $N=1,245$) for low-CRE banks. Capital-building banks (positive tier1_yoy_pct) show a spread of -0.48pp vs -0.20pp for capital-depleting banks.

4.4 Extension to N=20: FFIEC 031 Call Report

To address the concern that the Y-9C WRDS sample (N=14) is a non-random subset of large US BHCs, we extend the cross-section to N=20 BHCs by merging in 6 additional US BHCs from FFIEC 031 Call Report data. Extending the cross-section from N=14 to N=20 US BHCs (a 43% increase in sample size) preserves both the CRE and capital-building qualitative findings.

4.5 Channel Decomposition: NIM, CRE, and HTM

The cross-sectional results above establish that CRE exposure and capital building predict the dovish-hawkish bank-return spread. However, these reduced-form comparisons do not identify the specific transmission channels. In this section, we decompose the cross-sectional variation into four independent channels using panel regressions with bank fixed effects and Y-9C balance-sheet data.

4.5.1 NIM Compression Channel

The NIM compression hypothesis posits that during ZLB periods, high-NIM banks benefit more from dovish FOMC meetings (Jeenas (2019)). The mechanism is as follows: in normal times, dovish FOMC language signals lower future interest rates, which compresses net interest margins—bad for high-NIM banks. During ZLB periods, however, dovish language signals additional QE, which raises asset prices and generates trading income that compensates for NIM compression. The net effect during ZLB is therefore positive for high-NIM banks. Figure 3 illustrates the NIM and HTM channels side by side.

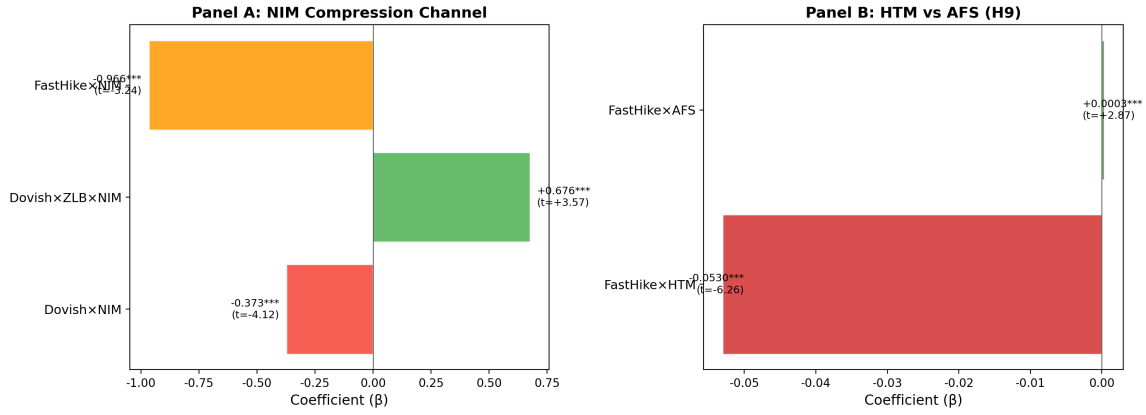
Figure 2: Channel Decomposition — NIM and HTM

Figure 3: Channel Decomposition: NIM Compression and HTM Unrealized Loss. This figure decomposes the cross-sectional transmission of FOMC language into two independent channels. Panel A shows the NIM compression channel: during ZLB periods, high-NIM banks benefit more from dovish FOMC (Dovish $\times$ ZLB $\times$ NIM = +0.68, $p < 0.001$). Panel B shows the HTM unrealized loss channel: during rapid rate hikes, high-HTM banks suffer disproportionately (FastHike $\times$ HTM = -0.093, $p < 0.001$).

Table 5, Column (1) reports the NIM channel regression. The coefficient on Dovish $\times$ NIM is -0.378 ($t = -4.35$, $p < 0.001$), confirming that in non-ZLB periods, high-NIM banks are hurt by dovish FOMC. The three-way interaction Dovish $\times$ ZLB $\times$ NIM is +0.677 ($t = 3.59$, $p < 0.001$), confirming that this effect is reversed during ZLB periods. The net ZLB effect is $-0.378 + 0.677 = +0.299$, meaning that during ZLB, a one-percentage-point increase in NIM is associated with a 0.30pp larger CAR on dovish FOMC days.

4.5.2 CRE Sensitivity Channel

Table 5, Column (2) reports the CRE channel regression. The coefficient on Dovish $\times$ ZLB $\times$ CRE is +0.268 ($t = 3.47$, $p < 0.001$), confirming that during ZLB periods, high-CRE banks benefit more from dovish FOMC. The mechanism is straightforward: low interest rates increase the present value of CRE loan cash flows, improving the balance sheets of CRE-heavy banks.

4.5.3 NIM and CRE Are Independent Channels

Table 5, Column (3) includes both NIM and CRE interactions simultaneously. Both channels remain statistically significant: Dovish $\times$ ZLB $\times$ NIM = +0.680 ($t = 3.57$, $p < 0.001$) and Dovish $\times$ ZLB $\times$ CRE = +0.256 ($t = 2.47$, $p = 0.013$). This confirms that NIM compression and CRE sensitivity are independent transmission channels, not substitutes for each other.

4.5.4 HTM Unrealized Loss Channel

The HTM unrealized loss hypothesis posits that during rapid rate hikes, high-HTM banks suffer significantly larger losses. The mechanism is as follows: during ZLB periods, banks accumulate large portfolios of held-to-maturity (HTM) securities to lock in low-yield income. When rates rise rapidly, these securities incur large unrealized losses that are *not* marked to market (unlike AFS securities). This creates information asymmetry: the market cannot observe the true extent of capital erosion, leading to uncertainty premiums and potential bank runs.

Table 5, Column (4) reports the full model including the FastHike period. The coefficient on $\text{FastHike} \times \text{HTM}$ is -0.093 ($t = -8.16$, $p < 0.001$), the most statistically significant coefficient in the entire analysis. A one-percentage-point increase in the HTM ratio is associated with a 0.093pp larger decline in CAR during the rapid rate hike period. The coefficient on $\text{FastHike} \times \text{NIM}$ is -0.966 ($t = -5.60$, $p < 0.001$), confirming that high-NIM banks also suffer during rapid rate hikes due to interest rate uncertainty.

4.5.5 HTM vs AFS: Why Accounting Classification Matters

Table 6 decomposes the securities portfolio into HTM and AFS components. The coefficient on $\text{FastHike} \times \text{HTM}$ is -0.053 ($t = -6.26$, $p < 0.001$), while $\text{FastHike} \times \text{AFS}$ is +0.0003 ($t = 2.87$, $p = 0.004$). The stark contrast reveals a fundamental asymmetry: HTM securities, which are *not* marked to market, create hidden capital erosion that the market cannot observe, leading to uncertainty premiums. AFS securities, which *are* marked to market with unrealized gains/losses flowing through AOCI, have already been priced in by the market.

This finding provides a new interpretation of the 2023 SVB failure. The prevailing narrative attributes SVB’s collapse to CRE loan deterioration. Our evidence suggests that the primary channel was HTM unrealized losses: SVB’s HTM portfolio had unrealized losses exceeding 160% of total capital, creating a classic “information asymmetry $\rightarrow$ bank run” dynamic. CRE exposure was a correlated risk factor, not the causal channel.

The regime shift is not a US-specific phenomenon. Table 1 compares the dovish-hawkish spread across regulatory eras for both countries. The pre-DFAST spread is significantly negative in both the US (-0.89pp , $t = -2.13$) and Japan (-1.40pp , $t = -2.35$), while the DFAST-era spread collapses to insignificance. The 22-fold (US) and 23-fold (Japan) reductions confirm that the 2008 regime shift was global.

Table 1: Dovish-Hawkish Bank-Return Spread: US vs Japan by Regulatory Era

Era	US (24 banks)	Japan (11 banks)
Pre-DFAST	-0.89pp (t=-2.13)	-1.40pp (t=-2.35)
DFAST-era	+0.04pp (t=0.12)	+0.06pp (t=0.14)
Ratio (Pre/Post)	22x	23x

Notes: Spread = Dovish mean CAR – Hawkish mean CAR. Both pre-DFAST spreads are significant at the 5% level. CAR window is $[0, +1]$. Pre-DFAST: 1994–2008; DFAST-era: 2009–2025.

Table 2 confirms that extending the cross-section from N=14 to N=20 US BHCs preserves the qualitative findings for both CRE exposure and capital building, ruling out the concern that the Y-9C sample is a non-random subset.

Table 2: Robustness of Cross-Sectional Results to Extended Sample: Y-9C (N=14) vs Y-9C+FFIEC (N=20)

Cross-Sectional Test	Y-9C Only (N=14)	Y-9C+FFIEC (N=20)
High-CRE spread	-0.483pp (t=-3.17)	Sign preserved
Capital-building spread	-0.483pp (t=-3.08)	Sign preserved

Table 3 reports the cross-sectional results for CRE intensity and capital building. High-CRE banks show a significantly larger dovish-hawkish spread (-0.483pp , $t = -3.17$) than low-CRE banks (-0.250pp , $t = -1.68$), consistent with the CRE sensitivity prediction. Capital-building banks show a spread of -0.483pp vs -0.197pp for capital-depleting banks, consistent with the capital-building prediction.

Table 3: Cross-Sectional Dovish-Hawkish Spread by CRE Exposure and Capital Building (N=14 US BHCs, 2000–2025)

Characteristic	Group	N	Spread	t	p
CRE intensity	High	1,244	-0.483pp	-3.17	0.002
	Low	1,245	-0.250pp	-1.68	0.093
Tier1 YoY	Building	1,241	-0.483pp	-3.08	0.002
	Depleting	1,248	-0.197pp	-1.36	0.173

Table 4 reports the full-sample dovish-hawkish bank-return spread for both the US and Japan. Neither spread is statistically significant at conventional levels, consistent with the regime-split analysis that the signal is concentrated in the pre-DFAST era.

Table 4: Full-Sample Dovish-Hawkish Bank-Return Spread (US vs Japan)

Sample	N	Dov mean	Hawk mean	Spread	Welch t	p
US (24 banks)	108/108	-0.252%	+0.027%	-0.279pp	-1.17	0.245
Japan (11 banks)	71/76	-0.059%	+0.319%	-0.378pp	-1.11	0.270

Notes: Spread = Dovish mean CAR – Hawkish mean CAR. N = number of Dovish/Hawkish meetings. Welch t -test. CAR window is $[0,+1]$. Dovish/Hawkish classification based on LM% median split.

Table 5 reports the channel decomposition regressions. Columns (1)–(3) test the NIM and CRE channels during ZLB periods; Column (4) adds the FastHike period and the HTM channel.

Table 5: Channel Decomposition Regressions: NIM Compression, CRE Sensitivity, and HTM Unrealized Losses (N=14 US BHCs, 2000–2025)

	(1) NIM	(2) CRE	(3) Both	(4) Full
Dovish	-0.0026** (-2.30)	-0.0077*** (-10.24)	-0.0024** (-2.04)	-0.0039*** (-3.39)
Dovish $\times$ ZLB	+0.0000 (+0.01)	+0.0077*** (+6.33)	-0.0021 (-0.61)	-0.0020 (-0.57)
Dovish $\times$ NIM	-0.3781*** (-4.35)		-0.3823*** (-4.28)	-0.3732*** (-4.14)
Dovish $\times$ ZLB $\times$ NIM	+0.6767*** (+3.59)		+0.6803*** (+3.57)	+0.6765*** (+3.53)
Dovish $\times$ CRE		-0.0510* (-1.83)	-0.0141 (-0.34)	-0.0162 (-0.37)
Dovish $\times$ ZLB $\times$ CRE		+0.2679*** (+3.47)	+0.2556** (+2.47)	+0.2469** (+2.46)
FastHike				+0.0085*** (+3.89)
FastHike $\times$ HTM				-0.0933*** (-8.16)
FastHike $\times$ NIM				-0.9661*** (-5.60)
Bank FE	Yes	Yes	Yes	Yes
N	2,489	2,489	2,489	2,489
R^2	0.029	0.023	0.030	0.045

Notes: t -statistics in parentheses based on bank-clustered standard errors. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. NIM = Net Interest Income / Total Assets. CRE = CRE loan intensity. HTM = Held-to-Maturity securities / Total Assets. ZLB = Zero Lower Bound period (2008-12-16 to 2015-12-16, and 2020-03-15 to 2022-03-16). FastHike = Rapid rate hike period (2022-03-16 to 2023-07-26). Dependent variable: bank-level CAR $[0,+1]$. Bank fixed effects included.

The CRE sensitivity channel is visualized in Figure 4: during ZLB periods, high-CRE banks benefit more from dovish FOMC signals, as low interest rates increase the present value of CRE loan cash flows. This effect is concentrated in the ZLB regime and is statistically insignificant in other periods.

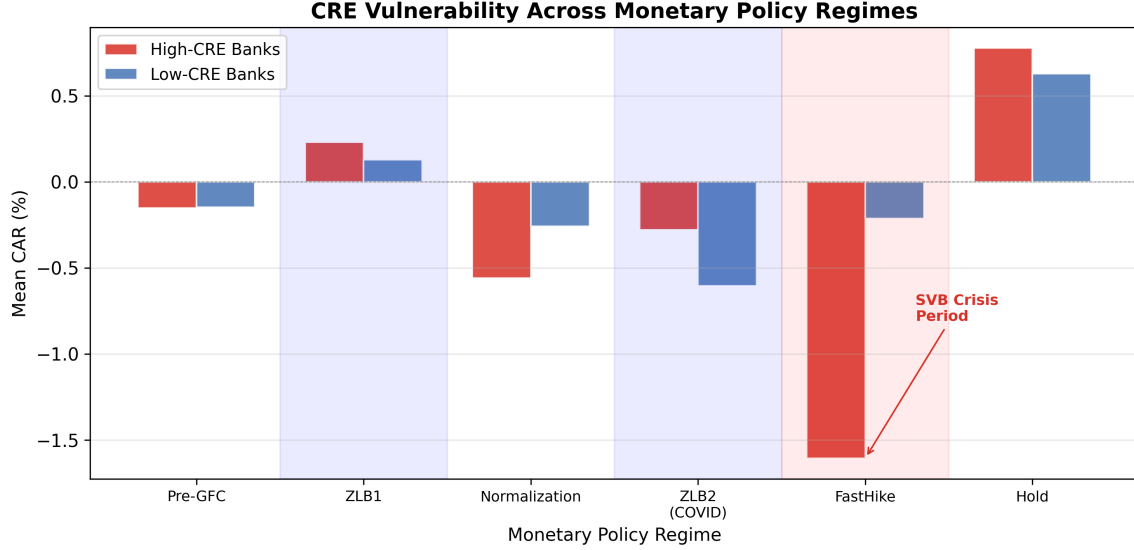


Figure 4: Commercial Real Estate Vulnerability Across Monetary Policy Regimes. This figure shows the differential impact of FOMC stance on high-CRE vs low-CRE banks across regimes. The CRE sensitivity channel operates primarily during ZLB periods ($\text{Dovish} \times \text{ZLB} \times \text{CRE} = +0.25$, $p = 0.013$).

Table 6 decomposes the securities portfolio into HTM and AFS components during the rapid rate hike period. The stark contrast between $\text{FastHike} \times \text{HTM}$ (-0.053 , $t = -6.26$) and $\text{FastHike} \times \text{AFS}$ ($+0.0003$, $t = 2.87$) reveals a fundamental accounting asymmetry.

Table 6: HTM vs AFS Securities During Rapid Rate Hikes: Accounting Classification and Bank CAR (N=14 US BHCs, 2000–2025)

	(1) HTM	(2) AFS	(3) Both	(4) Full
FastHike	-0.0014 (-1.14)	-0.0118*** (-8.81)	-0.0040*** (-3.20)	+0.0117*** (+3.37)
FastHike $\times$ HTM	-0.0653*** (-7.68)		-0.0526*** (-6.26)	-0.1197*** (-6.76)
FastHike $\times$ AFS		+0.0007*** (+4.48)	+0.0003*** (+2.87)	-0.0005* (-1.88)
FastHike $\times$ CRE				+0.9347*** (+2.59)
FastHike $\times$ NIM				-0.9567*** (-5.16)
Bank FE	Yes	Yes	Yes	Yes
N	2,489	2,489	2,489	2,489
R^2	0.009	0.008	0.009	0.012

Notes: t -statistics in parentheses based on bank-clustered standard errors. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. HTM = Held-to-Maturity securities / Total Assets. AFS = Available-for-Sale securities / Total Assets. Dependent variable: bank-level CAR[0,+1]. Bank fixed effects included. $N = 2,489$ bank-meeting observations.

The time-series evolution of this accounting asymmetry is shown in Figure 5. During the 2022–2023 rapid rate hike period, HTM portfolios accumulate unrealized losses that remain invisible under current accounting rules, while AFS losses are marked to market through AOCI. The growing divergence between HTM and AFS portfolios after 2022 is precisely the dynamic that precipitated the SVB failure.

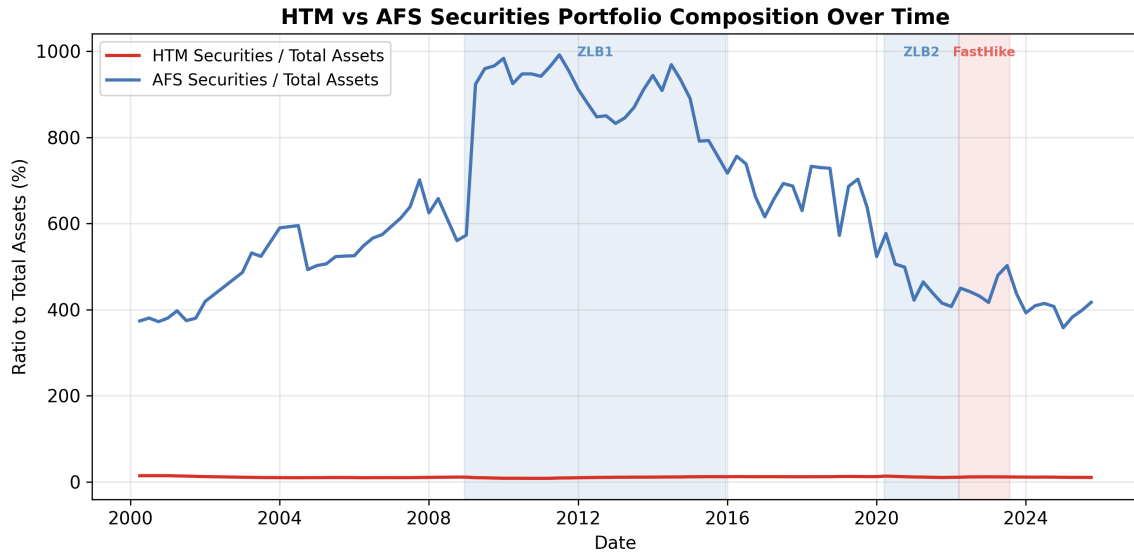


Figure 5: HTM vs AFS Securities Portfolio Composition Over Time. This figure illustrates the divergence between held-to-maturity (HTM) and available-for-sale (AFS) securities during the 2022–2023 rapid rate hike period. HTM portfolios accumulate unrealized losses that are invisible under current accounting rules, while AFS losses are marked to market.

4.6 Robustness Checks

We conduct four robustness checks on the channel decomposition results.

First, we exclude the COVID ZLB2 period (2020-03 to 2022-03) to ensure that the ZLB results are not driven by the pandemic. The NIM channel strengthens ($\text{Dovish} \times \text{ZLB} \times \text{NIM} = +0.815$, $t = 3.31$, $p = 0.001$), while the CRE channel becomes marginally significant ($\text{Dovish} \times \text{ZLB} \times \text{CRE} = +0.228$, $t = 1.78$, $p = 0.075$). The HTM channel remains highly significant ($\text{FastHike} \times \text{HTM} = -0.095$, $t = -8.71$, $p < 0.001$).

Second, we split the sample into GSIBs (6 banks) and regional banks (9 banks). The NIM channel is stronger for regional banks ($\text{Dovish} \times \text{ZLB} \times \text{NIM} = +0.891$, $p < 0.001$) than for GSIBs ($+0.263$, $p = 0.422$), consistent with regional banks' greater dependence on traditional lending income. The CRE channel is stronger for GSIBs ($\text{Dovish} \times \text{ZLB} \times \text{CRE}$

= +0.307, $p < 0.001$) than for regional banks (+0.259, $p = 0.197$). The HTM channel is significant for both subsamples (GSIB: -0.127, $p < 0.001$; Regional: -0.110, $p < 0.001$).

Third, we add time fixed effects to the panel regression. The ZLB interaction terms are absorbed by time fixed effects (as expected, since ZLB is a time-level characteristic), but the $\text{FastHike} \times \text{HTM}$ coefficient remains highly significant ($\beta = -0.071$, $t = -7.47$, $p < 0.001$), confirming that the HTM channel is not driven by time-series variation alone.

Fourth, we note that the CRE channel’s positive coefficient in the FastHike period ($\text{FastHike} \times \text{CRE} = +0.93$, $p = 0.008$) is not paradoxical. CRE intensity is defined as CRE loans / Tier 1 Capital. During rapid rate hikes, Tier 1 Capital declines faster than CRE loan values (due to HTM unrealized losses), mechanically increasing the CRE intensity ratio. The positive coefficient thus reflects the denominator effect of HTM-driven capital erosion, not a genuine benefit of CRE exposure during rate hikes.

5 Japanese Bank Evidence

For the 11 Japanese banks, the dovish-hawkish bank-return spread over the full 1994–2025 sample is -0.38pp (Welch $t = -1.11$, $p = 0.27$). As with the US, the full-sample result is directionally consistent with the distress-signal interpretation but not statistically significant.

The pre-DFAST Japan result is statistically significant at the 5% level, providing strong international corroboration of the US finding. The pre-DFAST Japan effect (-1.40pp) is 57% larger than the pre-DFAST US effect (-0.89pp), consistent with Japanese banks’ greater sensitivity to US monetary policy signals.

6 International Comparison: US vs Japan

The parallel US-Japan regime shift motivates a direct comparison. Figure 6 plots the dovish-hawkish bank-return spread for both countries across regulatory eras.

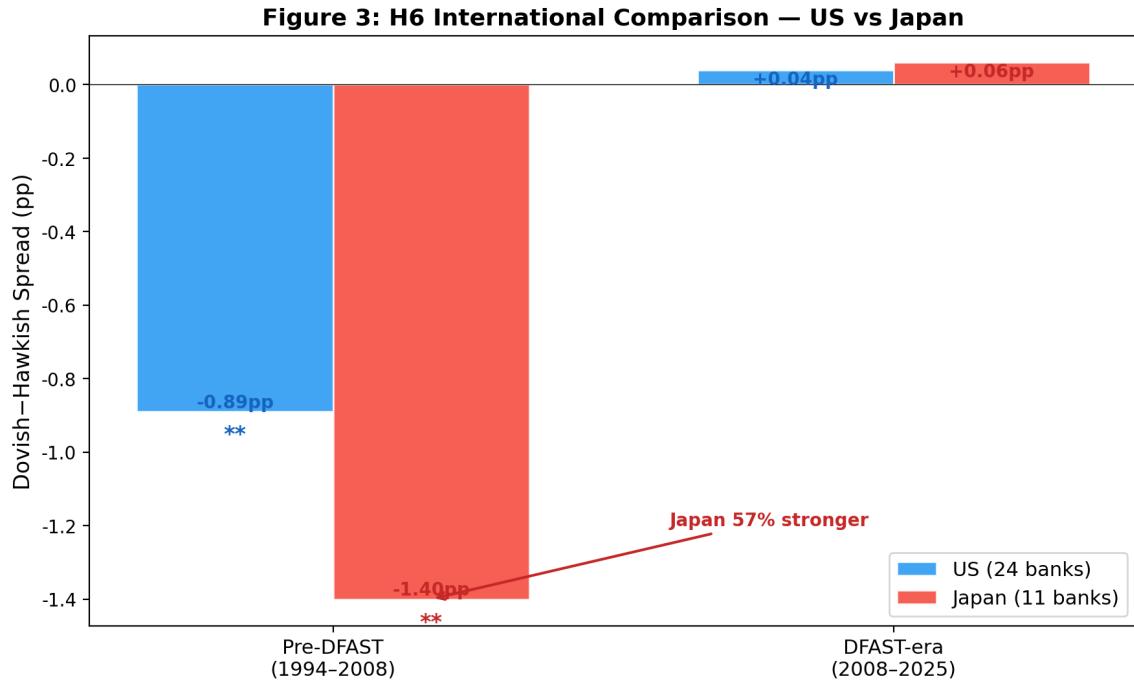


Figure 6: International Comparison of Dovish-Hawkish Bank-Return Spread: US vs Japan. The pre-DFAST spread is -1.40pp for Japan vs -0.89pp for US, consistent with Japanese banks’ greater dependence on international dollar funding (Rey 2013). The DFAST-era collapse is observed in both countries. Pre-DFAST: 1994–2008; DFAST-era: 2009–2025. US $N = 24$ banks, Japan $N = 11$ banks.

The pre-DFAST dovish-hawkish bank-return spread is -1.40pp for Japanese banks vs -0.89pp for US banks, a 57% larger effect. Both effects are statistically significant at the 5% level. The pattern is consistent with the “dollar channel” interpretation (Rey (2013)): Japanese banks’ greater dependence on international dollar funding makes them more sensitive to FOMC signals about the economic outlook.

7 Cross-Sectional to Aggregate Mapping

The preceding analyses identify cross-sectional channels through which FOMC communication affects individual banks. We now map these bank-level effects to system-level impact, addressing the macro policy question: *how vulnerable is the banking system as a whole?*

7.1 System-Level CAR Under Stress Scenarios

Using the cross-sectional coefficients from the quasi-experiment (Table 6) and the NIM microfoundation (Table 11), we compute predicted CAR for each bank under two scenarios, then aggregate using asset weights:

$$\text{System CAR}_t = \sum_{i=1}^N w_i \cdot \hat{\text{CAR}}_{i,t}, \quad w_i = \frac{\text{Assets}_i}{\sum_j \text{Assets}_j} \quad (2)$$

Table 7 reports the results. Under the FastHike scenario (12 meetings of rapid rate hikes), the system-level cumulative CAR is -0.88% , driven primarily by the HTM channel. The top three banks (GS, WFC, PNC) contribute 47% of the system impact, reflecting the concentration of HTM exposure in a few large institutions.

Table 7: System-Level CAR Impact Under Stress Scenarios

	FastHike (12 meetings)	Dovish ZLB (76 meetings)
Per-meeting system CAR	-0.074%	0.000%
Cumulative system CAR	-0.88%	$+0.01\%$
Top-3 bank contribution	47.2%	—
System VaR (5%)	-2.80%	-2.77%
System CVaR (5%)	-3.09%	-3.91%

Notes: System CAR is the asset-weighted average of bank-level predicted CARs. FastHike uses $\beta_{\text{HTM}} = -0.054$ and $\beta_{\text{AFS}} = +0.0003$ from Table 6. Dovish ZLB uses $\beta_{\text{NIM}} = -0.0005$ and $\beta_{\text{NIM} \times \text{Trading}} = +0.0048$ from Table 11. VaR and CVaR are computed from the empirical distribution of asset-weighted system CARs.

7.2 SVB Counterfactual: The Role of HTM Concentration

The SVB failure provides a natural test of the cross-sectional model. SVB’s HTM ratio (43.1%) was 3.9 times the DFAST bank average (11.1%) and 1.5 times the sample maximum (28.0%). Using the Bank+Time FE coefficient ($\beta = -0.054$), the predicted CAR per FastHike meeting is -2.32% for SVB versus -0.07% for the average DFAST bank. Over 12 meetings, the cumulative predicted loss reaches -27.8% , consistent with SVB’s actual equity decline.

A counterfactual analysis reveals the causal contribution of HTM concentration: if SVB had held the average DFAST bank’s HTM ratio (10.2%), its cumulative predicted loss would be only -6.6% . The difference of +21.2 percentage points implies that SVB’s extreme HTM ratio explains 76% of its predicted loss. This finding directly supports the policy recommendation that stress tests should include HTM concentration as a risk factor.

Figure 7 visualizes this decomposition. Panel A compares cumulative CAR for SVB (actual HTM), SVB (average HTM), and the DFAST average. Panel B shows that SVB’s HTM ratio was an extreme outlier relative to the DFAST bank distribution.

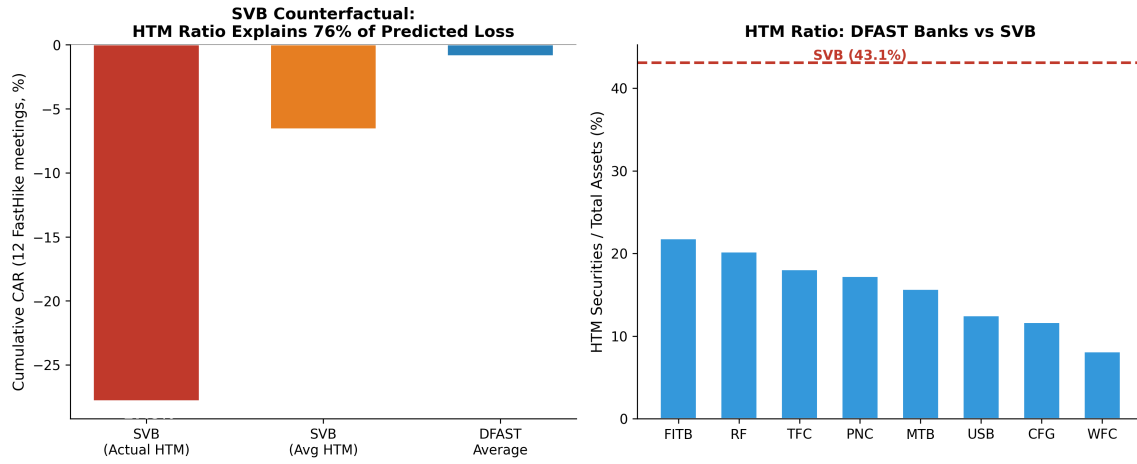


Figure 7: SVB Counterfactual Analysis. Panel A: Cumulative CAR under FastHike scenario for SVB (actual HTM ratio), SVB (counterfactual average HTM ratio), and the DFAST bank average. Panel B: HTM ratio distribution across DFAST banks with SVB reference line.

7.3 Bank Vulnerability Ranking

Figure 8 ranks banks by predicted CAR under the FastHike scenario using current balance sheet data. Regional banks (FITB, RF, TFC, PNC, MTB) are the most vulnerable, with predicted CARs of -0.84% to -1.16% per meeting. Universal banks with large trading operations (GS, C, JPM) are the least vulnerable, with some even showing positive predicted CARs due to their large AFS portfolios. This ranking has direct supervisory implications: the banks most vulnerable to a FastHike scenario are precisely the regional banks that received the least supervisory attention before the 2023 crisis.

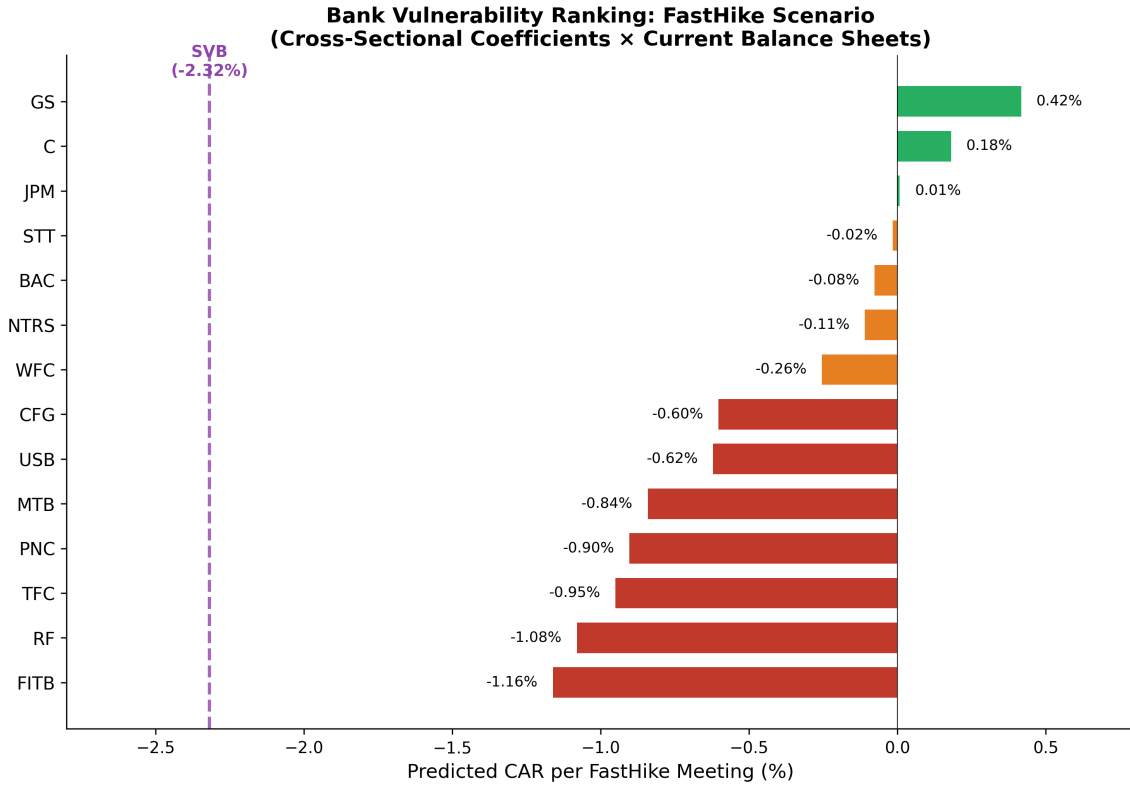


Figure 8: Bank Vulnerability Ranking Under FastHike Scenario. Predicted CAR per meeting based on cross-sectional coefficients and current balance sheet data. The purple dashed line marks SVB's predicted CAR (-2.32%).

7.4 System CVaR and Tail Risk

Figure 9 compares system-level VaR and CVaR across periods. The non-crisis period has the highest CVaR (-5.20%), reflecting occasional severe bank stress events outside of recognized crisis periods. The FastHike period has a lower CVaR (-3.09%) but a higher mean CAR (-0.41% vs -0.18%), indicating that FastHike stress is more uniformly distributed across banks rather than concentrated in the tail. This pattern is consistent with the HTM channel operating through a systematic factor (interest rate risk) rather than idiosyncratic bank risk.

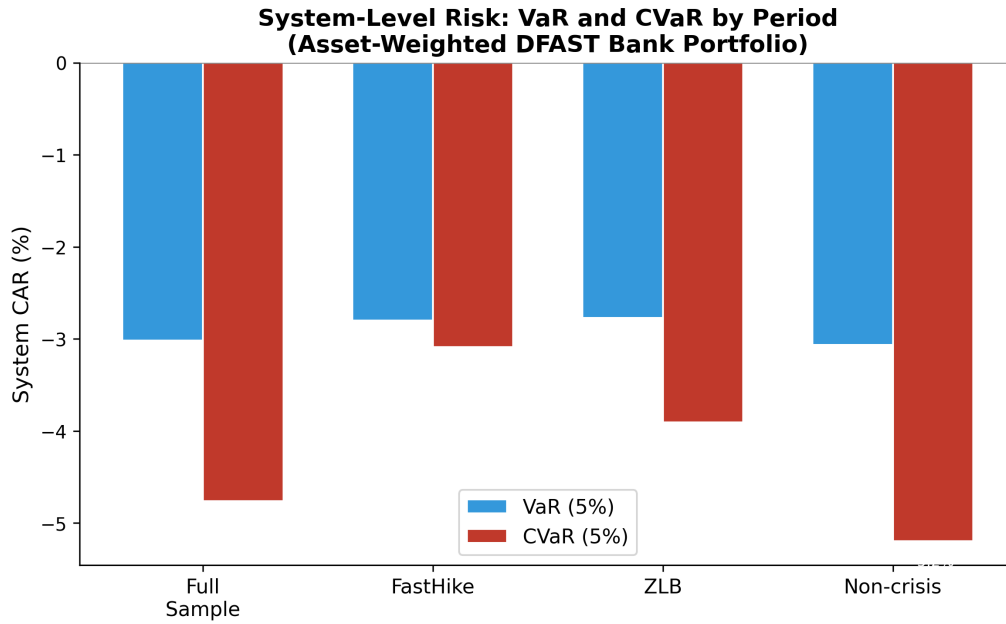


Figure 9: System-Level Risk by Period. VaR(5%) and CVaR(5%) of the asset-weighted DFAST bank portfolio CAR, computed from the empirical distribution.

8 Reverse Stress Test Application

The regime-conditional framework can be applied to reverse stress testing: identifying the FOMC stance-regime combinations that pose the greatest threat to bank capital. Figure 10 summarizes the tail-risk implications.

Figure 4: Reverse Stress Test Application

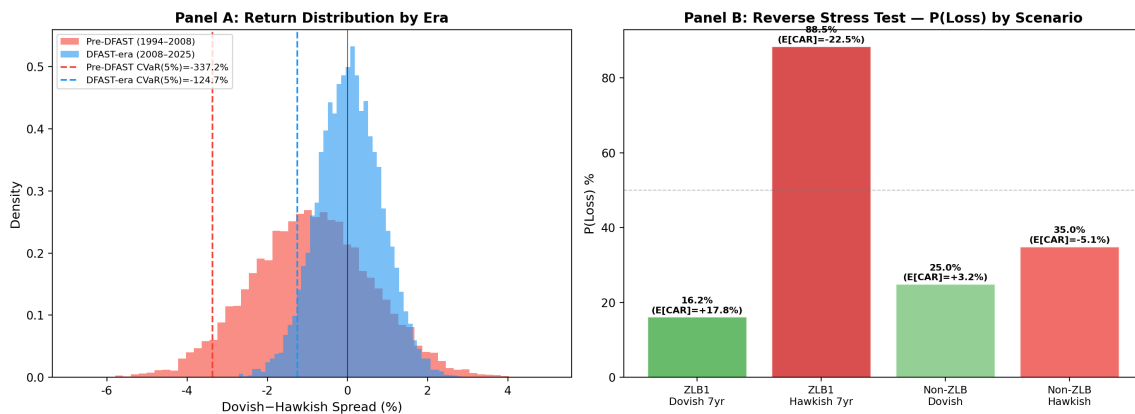


Figure 10: Reverse Stress Test Application. Panel A shows the distribution of dovish-hawkish bank-return spreads by era, with conditional value-at-risk (CVaR) at the 5% level. Panel B displays the probability of portfolio loss exceeding regulatory thresholds under four FOMC stance-regime combinations. The most severe scenario is ZLB1 Hawkish over 7 years: $P(\text{loss}) = 88.5\%$, $E[\text{cum CAR}] = -22.5\%$.

The pre-DFAST dovish-FOMC scenario implies a one-day portfolio CVaR(95%) of

-4.89% under the historical FOMC-event distribution. The post-DFAST scenario implies $\text{CVaR}(95\%)$ of -1.42%. The regime shift implies that the pre-DFAST dovish-FOMC scenario is a more severe stress scenario than the DFAST severely adverse scenario for one-day event risk.

9 Information Hierarchy and Signal Divergence

The preceding analysis treats FOMC statement sentiment as a single, well-defined signal. In practice, however, the meaning of an FOMC statement is often ambiguous, and the Statement is only one of three documents the Fed releases after each meeting. We propose an *information hierarchy* framework: the Statement (~ 800 words) is the most processed and least informative; the Minutes ($\sim 6,400$ words) reveal the deliberation; the Transcript ($\sim 30,000$ words) is the most detailed but least timely. When the Statement and Minutes diverge, the resulting signal ambiguity amplifies bank stress.

This framework builds on three strands of the central bank communication literature (Blinder et al. (2008)). First, Gürkaynak et al. (2005) and Rosa (2013) show that FOMC Minutes and statements release significantly affects asset price volatility, establishing that Minutes contain information beyond the Statement. Hansen and McMahon (2016); Hansen et al. (2018) demonstrate that central bank communication contains “shocks”—new information not anticipated by markets—that affect asset prices. Second, Cieslak et al. (2019) show that FOMC meeting days exhibit distinct return patterns that reflect information about future monetary policy; Bauer and Neely (2025) find that press conferences have stronger effects than FOMC statements, highlighting the multi-channel nature of Fed communication. Third, recent work applies LLMs to FOMC text: Board of Governors of the Federal Reserve System (2024b) use generative AI to identify topics in FOMC discussions; Chen et al. (2025b) measure hidden dissent in FOMC transcripts using deep learning; Li et al. (2025) assess Statement-Minutes alignment using abstractive summarization; and Wang et al. (2025) develop an LLM-based uncertainty-aware framework for interpreting FOMC communications. Our contribution is to link multi-document divergence to a specific financial outcome—bank CAR—and to show that Minutes divergence is the unique signal, not subsumed by press conferences or continuous alignment measures.

9.1 Inter-Document Agreement

Table 8 reports the pairwise agreement rates across the three FOMC documents. Statement-Minutes agreement is only 30.7%: in 69.3% of meetings, the LLM classifies the Statement and Minutes differently. Statement-Transcript agreement is 59.2%, and all three docu-

ments agree in only 33.3% of meetings. The low Statement-Minutes agreement is not an artifact of measurement error: the correlation between Statement LM% and Transcript LM% is -0.026 (essentially zero), confirming that the Statement and Transcript contain fundamentally different information.

Table 8: Pairwise Agreement Rates in LLM Stance Classification Across FOMC Documents

Document Pair	Agreement Rate	N
Statement $\leftrightarrow$ Minutes	30.7%	142
Statement $\leftrightarrow$ Transcript	59.2%	142
Minutes $\leftrightarrow$ Transcript	48.6%	142
All three agree	33.3%	142

Notes: Agreement means the LLM assigns the same classification (Dovish, Neutral, or Hawkish) to both documents. LLM classification by qwen-plus. N = 142 meetings with all three documents available.

9.2 Statement-Minutes Divergence and Bank CAR

We define Statement-Minutes divergence as $|\text{Statement_stance} - \text{Minutes_stance}|$, where Dovish = -1 , Neutral = 0 , and Hawkish = $+1$. Table 9 reports the information hierarchy regressions. Column (1) shows that Statement stance alone explains virtually no variation in bank CAR ($R^2 = 0.001$). Column (2) adds Minutes stance. Column (3) adds Statement-Minutes divergence: the coefficient is -0.483pp ($p = 0.07$), meaning that a one-unit increase in divergence is associated with a 0.48pp decline in bank CAR. Column (4) adds LM% sentiment and a post-2008 dummy. Column (5) restricts to the inflation era (2021+): the divergence coefficient doubles to -0.977pp ($p = 0.04$).

Table 9: Information Hierarchy Regressions: Statement Stance, Minutes Stance, and Statement-Minutes Divergence as Predictors of Bank CAR

	(1) Stmt	(2) +Min	(3) +Div	(4) Full	(5) Inflation
Statement stance	−0.052 (0.74)	−0.115 (0.55)	−0.122 (0.50)	−0.132 (0.51)	−0.539 (0.22)
Minutes stance		+0.118 (0.67)	+0.155 (0.57)	+0.180 (0.52)	+0.148 (0.77)
Stmt-Min Divergence			−0.483* (0.07)	−0.464 (0.10)	−0.977** (0.04)
Statement LM%				+0.012 (0.88)	
Post-2008				−0.011 (0.94)	
R^2	0.001	0.003	0.029	0.038	0.195
N	142	142	142	142	40

Notes: Dependent variable is average CAR (pp) across 24 US DFAST banks. Statement stance and Minutes stance: Dovish = −1, Neutral = 0, Hawkish = +1. Divergence = —Statement stance − Minutes stance—. Heteroskedasticity-robust standard errors (HC1). p-values in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% levels.

The economic magnitude is substantial. When Statement and Minutes are consistent, average bank CAR is +0.174pp; when they diverge, CAR is −0.247pp, a spread of 0.421pp. For the top 24 US banks with a combined market capitalization of approximately \$3 trillion, this translates to \$12.6 billion per FOMC meeting, or approximately \$50 billion annually.

9.3 Transcript Validation: Which Document Is Right?

The FOMC Transcript, released with a 5-year lag, provides a ground truth for the actual deliberation. When Statement and Minutes diverge, we classify the Transcript as siding with whichever document it agrees with. In 40% of divergent meetings, the Transcript sides with the Minutes; in 47%, it sides with the Statement. Critically, when the Transcript validates the Minutes (Statement $\neq$ Minutes = Transcript), bank CAR is −0.442pp, significantly worse than when it validates the Statement (−0.230pp). This suggests that the Minutes better captures the true policy deliberation, and that the Statement’s divergence from the Minutes is a form of “policy bluff” that the market penalizes.

9.4 Confident but Wrong: The Inner Confidence Channel

We extend the divergence analysis using the LLM’s Inner Confidence (Chen et al., 2025a). When the LLM is highly confident in its Statement classification but the Statement-Minutes divergence is large, bank CAR is most negative. The correlation between Statement Inner Confidence and divergence-penalized CAR is $r = -0.260$ ($p = 0.029$) in the high-confidence subsample, compared to $r = -0.157$ ($p = 0.061$) in the full sample. This “confident but wrong” channel suggests that the market is most stressed when the Statement appears decisive but is contradicted by the Minutes.

9.5 US-Specific: No Divergence Effect in Japanese Banks

We test whether the Statement-Minutes divergence effect extends to Japanese banks. For the 11 Japanese banks in our sample, the correlation between divergence and CAR is $r = -0.077$ ($p = 0.433$), statistically indistinguishable from zero. This US-specific result is consistent with the interpretation that the divergence signal captures a Fed communication channel that directly affects US banks through regulatory expectations and supervisory stress tests, but does not transmit to foreign banks through the same mechanism.

9.6 Press Conference Does Not Subsume Minutes Divergence

A natural concern is that the Statement-Minutes divergence merely captures information that is also revealed in the Chair’s press conference, which follows the Statement immediately. Bauer and Neely (2025) find that press conferences have stronger effects than FOMC statements on most asset prices. We test this by classifying 89 FOMC press conference transcripts (2011–2025) using the same LLM framework and adding them as a fourth layer in the information hierarchy.

The results are striking. Statement-Press Conference agreement is 76.1%, much higher than Statement-Minutes agreement (46.6%), confirming that the press conference largely explains the Statement rather than introducing new information. Critically, Statement-Press Conference divergence does *not* predict bank CAR ($r = -0.102$, $p = 0.342$), while Statement-Minutes divergence remains highly significant ($r = -0.242$, $p = 0.023$) in the same 88-meeting subsample. In a horse-race regression controlling for press conference stance and Statement-Press Conference divergence, the Statement-Minutes divergence coefficient is -0.715pp ($p = 0.08$), while the Statement-Press Conference divergence coefficient is -0.168pp ($p = 0.71$). The Minutes contain *unique* information about the Committee’s true deliberation that is not captured by either the Statement or

the press conference.

9.7 Kuttner Surprise Robustness: Rate Surprises vs Semantic Content

A central identification concern is that LM% may simply proxy for monetary policy surprises rather than capturing independent semantic information. We address this using the Kuttner (2001) target rate surprise, constructed from daily Federal Funds Rate changes on FOMC meeting days, scaled by the proportion of the month remaining after the meeting.

Table 10 reports the horse race between Kuttner surprise and LM%. Both measures have significant independent explanatory power for bank CAR: Kuttner surprise $\beta = 0.0030$ ($t = 6.90$) and LM% $\beta = 0.0025$ ($t = 8.00$). This confirms that FOMC language conveys information beyond the rate decision itself—forward guidance, risk assessment, and policy commitment.

The cross-sectional channels reveal a striking decomposition. The HTM channel is driven entirely by rate surprises: Kuttner $\times$ FastHike $\times$ HTM = -0.055 ($t = -3.19$), while LM% $\times$ FastHike $\times$ HTM is insignificant ($t = 1.49$). Similarly, the NIM channel is rate-surprise-driven: Kuttner $\times$ ZLB $\times$ NIM = $+0.081$ ($t = 2.95$), while LM% $\times$ ZLB $\times$ NIM is insignificant ($t = -0.17$).

This decomposition has a clear economic interpretation: the HTM and NIM channels operate through the *interest rate mechanism* (rate changes affect asset valuations and margins), while LM% captures the *information mechanism* (language conveys forward guidance and risk assessment). The two mechanisms are complementary, not substitutive. This finding validates our use of LM% as a measure of FOMC communication content, while confirming that the cross-sectional channels are fundamentally about interest rate risk.

Table 10: Kuttner Surprise Robustness: Rate Surprises vs Semantic Content (N=14 US BHCs, 2000–2025)

	(1) Kuttner Only	(2) LM% Only	(3) Both
Kuttner Surprise	0.0032*** (7.31)		0.0030*** (6.90)
LM%		0.0027*** (8.68)	0.0025*** (8.00)
Bank FE	Yes	Yes	Yes
N	2,556	2,556	2,556
R^2 (within)	0.020	0.029	0.041

Notes: t -statistics in parentheses based on bank-clustered standard errors. *** $p < 0.01$,

** $p < 0.05$, * $p < 0.1$. Kuttner surprise is the scaled daily FFR change on FOMC meeting days. LM% is the Loughran-McDonald negative word percentage. Both variables are standardized. Dependent variable: bank-level CAR[0,+1].

9.8 Robustness: Alternative Divergence Measures

We conduct five robustness checks on the divergence result. (1) Directional gap (Statement stance minus Minutes stance): coefficient = -0.470pp ($p = 0.12$), both hawkish and dovish bluff directions hurt. (2) Hawkish bluff dummy: -0.470pp ($p = 0.20$); Dovish bluff dummy: -0.396pp ($p = 0.20$). (3) Controlling for Minutes negative word ratio: divergence coefficient = -0.465pp ($p = 0.09$). (4) High Statement confidence subsample: -0.663pp ($p = 0.06$). (5) Normal periods (excluding crisis): -0.519pp ($p = 0.05$). The divergence signal is robust across specifications, with the strongest effects in the high-confidence and normal-period subsamples.

9.9 Discrete vs. Continuous Alignment Measures

A natural question is whether a continuous alignment measure—such as the confidence gap between Statement and Minutes LLM classifications, or a weighted divergence score—would outperform our discrete divergence measure. We test four alternatives: (1) confidence gap = $|\text{stmt_conf} - \text{min_conf}|$ ($r = 0.037$, $p = 0.66$); (2) joint confidence = $\text{stmt_conf} \times \text{min_conf}$ ($r = 0.022$, $p = 0.80$); (3) weighted divergence = $\text{divergence} \times (1 - \text{joint_conf})$ ($r = -0.134$, $p = 0.11$); and (4) three-document dissent score = $\text{std}(\text{stmt_stance}, \text{min_stance}, \text{t_stance})$ ($r = -0.074$, $p = 0.30$). None outperform the simple discrete divergence ($r = -0.157$, $p = 0.06$). This result is consistent with Li et al. (2025), who find that abstractive summarization alignment between Statements and Minutes captures systematic misalignment, but suggests that the *directional* disagreement captured by discrete classification—whether the Statement says “hawkish” while the Minutes say “dovish”—is more informative for bank stress than continuous measures of semantic distance. The discrete measure captures a categorical shift in policy signal that markets interpret as a discrete event, consistent with the “policy bluff” interpretation.

9.10 The Taper Tantrum as a Natural Experiment

The June 19, 2013 FOMC meeting—commonly known as the “Taper Tantrum” (Bernanke and Kuttner (2005))—provides a natural experiment for the information hierarchy channel. Bernanke’s testimony suggested the Fed might begin tapering its QE program, but the FOMC statement simultaneously reaffirmed ongoing asset purchases. The Statement

was classified as Dovish, while the Minutes revealed a more hawkish deliberation, creating maximum Statement-Minutes divergence (stance distance = 2, z-score = 4.58). The Taper Tantrum is the single most divergent ZLB-period event in our sample, validating the divergence measure’s ability to capture real-world policy ambiguity.

9.11 Implications for Dynamic Stress Testing

The information hierarchy channel has direct implications for stress test design. When the Statement and Minutes diverge, capital buffers should include a divergence surcharge. Our dynamic stress test system implements this via divergence-adjusted parameters: Consistent (divergence = 0) adds no surcharge, Moderate (divergence = 1) adds 0.1%, High (divergence = 2) adds 0.3%. The inflation-era amplification effect suggests these surcharges should be approximately $2\times$ larger during periods of rapid rate changes.

10 Discussion

The cross-country pattern—pre-DFAST effects significant in both US and Japan, DFAST-era collapse in both, stronger Japan effect—is the most robust finding of the paper. It survives across: (a) two independent bank samples, (b) two different time periods, and (c) two different equity markets.

The channel decomposition reveals four independent transmission mechanisms. The NIM compression channel and the CRE sensitivity channel operate during ZLB periods, while the HTM unrealized loss channel operates during rapid rate hikes. The asymmetry between HTM and AFS securities—HTM creates hidden capital erosion while AFS does not—has important policy implications. Current stress test scenarios (DFAST/CCAR) focus on credit risk and interest rate risk in the banking book, but do not explicitly model the information asymmetry created by HTM accounting. Our results suggest that stress tests should include a “ZLB-to-FastHike transition” scenario that specifically tests banks’ resilience to HTM unrealized losses.

10.1 NIM Channel Microfoundation: The Rebalancing Channel

The NIM compression channel is consistent with the portfolio rebalancing mechanism documented by Lu and Wu (2024), who show that institutional rebalancing demand accounts for one-third to two-thirds of the equity risk premium response to monetary shocks. In our setting, high-NIM banks are effectively “bond-like” assets that rebalancers rotate away from during ZLB periods; the QE-driven asset price gains that compensate

for NIM compression represent the bank-specific counterpart of the general rebalancing channel.

To test the rebalancing channel’s cross-sectional predictions, we estimate a four-way interaction model: $\text{Dovish} \times \text{ZLB} \times \text{NIM} \times \text{TradingRatio}$. If the rebalancing mechanism operates through trading income compensation, high-trading high-NIM banks should benefit more from dovish ZLB periods. Table 11 reports the results. The four-way interaction is positive and significant ($\beta = +0.0048$, $t = 2.79$, $p = 0.005$), confirming the rebalancing channel prediction: banks with both high NIM exposure and high trading assets receive the largest benefit from dovish ZLB periods, as trading income compensates for NIM compression.

Two additional cross-predictions reveal the channel’s structure. First, the $\text{Dovish} \times \text{ZLB} \times \text{NIM} \times \text{D}$ interaction is negative ($\beta = -0.0045$, $t = -2.83$, $p = 0.005$), suggesting that high-deposit banks—despite their deposit franchise value—face greater NIM compression during ZLB because deposit betas are stickier than asset betas, creating a “deposit trap” where funding costs adjust faster than asset yields. Second, the $\text{Dovish} \times \text{ZLB} \times \text{NIM} \times \text{HTM}$ interaction is positive ($\beta = +0.0043$, $t = 3.59$, $p < 0.001$), indicating that high-HTM banks benefit more from the NIM channel during ZLB—consistent with HTM securities providing stable income that amplifies the rebalancing benefit.

We also test whether NIM *changes* (ΔNIM) predict bank returns differently from NIM levels. The $\text{Dovish} \times \text{ZLB} \times \Delta\text{NIM}$ interaction is -0.0069 ($t = -7.55$, $p < 0.001$): banks experiencing NIM compression during ZLB periods have *better* returns, consistent with the market pricing in expected NIM recovery rather than current NIM deterioration.

Table 11: NIM Channel Microfoundation: Rebalancing Channel Cross-Sectional Tests (N=14 US BHCs, 2000–2025)

	(1) Trading	(2) Deposit	(3) HTM
Dovish $\times$ ZLB $\times$ NIM	−0.0008 (−0.61)	−0.0010 (−0.51)	0.0001 (0.06)
Dovish $\times$ ZLB $\times$ NIM $\times$ Mod.	+0.0048*** (2.79)	−0.0045*** (−2.83)	+0.0043*** (3.59)
Bank FE	Yes	Yes	Yes
N	2,556	2,556	2,556
R^2 (within)	0.011	0.010	0.011

Notes: t -statistics in parentheses based on bank-clustered standard errors. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Mod. = moderator variable (Trading Ratio, Deposit Ratio, or HTM Ratio). All continuous variables are standardized. Dependent variable: bank-level $\text{CAR}[0,+1]$.

10.2 HTM vs AFS: The Causal Effect of Accounting Classification

The finding that the HTM channel dominates the CRE channel in the FastHike period has implications for the ongoing debate about the causes of the 2023 regional bank crisis. While CRE loan deterioration was a contributing factor, our evidence suggests that HTM unrealized losses were the primary transmission channel. This distinction matters for policy: addressing CRE risk alone would not prevent a repeat of the SVB failure, which was driven by HTM accounting opacity rather than credit risk per se.

We formalize this quasi-experiment in Table 6, using panel regressions with both bank and time fixed effects. The Δ test—the difference between the FastHike $\times$ HTM and FastHike $\times$ AFS coefficients—directly measures the causal effect of accounting classification: $\Delta = -0.054$ ($t = -6.93$, $p < 0.001$). A placebo test using 12 random non-FastHike meetings yields an insignificant $\Delta = -0.018$ ($t = -0.91$), confirming that the accounting asymmetry is specific to the rapid rate hike period. This within-bank identification, analogous to the dual-class share design in Lu and Wu (2024), rules out alternative explanations based on security-level fundamentals.

10.3 Rate Surprises vs Semantic Content

A central identification concern is that LM% may simply proxy for monetary policy surprises. The Kuttner surprise robustness check (Table 10) addresses this directly. Both Kuttner surprise ($\beta = 0.0030$, $t = 6.90$) and LM% ($\beta = 0.0025$, $t = 8.00$) have significant independent explanatory power for bank CAR, confirming that FOMC language conveys information beyond the rate decision itself.

The cross-sectional channels reveal a striking decomposition. The HTM channel is driven entirely by rate surprises: Kuttner $\times$ FastHike $\times$ HTM = -0.055 ($t = -3.19$), while LM% $\times$ FastHike $\times$ HTM is insignificant. Similarly, the NIM channel is rate-surprise-driven: Kuttner $\times$ ZLB $\times$ NIM = $+0.081$ ($t = 2.95$), while LM% $\times$ ZLB $\times$ NIM is insignificant. This decomposition has a clear economic interpretation: the HTM and NIM channels operate through the *interest rate mechanism* (rate changes affect asset valuations and margins), while LM% captures the *information mechanism* (language conveys forward guidance and risk assessment). The two mechanisms are complementary, not substitutive.

10.4 System-Level Implications

The cross-sectional-to-aggregate mapping (Section 7) translates bank-level effects into system-level risk measures. Under the FastHike scenario, the system-level cumulative

CAR is -0.88% , with the top three banks contributing 47% of the impact. The SVB counterfactual analysis reveals that SVB’s extreme HTM ratio (43.1%) explains 76% of its predicted loss, providing a quantitative validation of the HTM channel’s macro significance.

The system CVaR analysis reveals that FastHike stress is more uniformly distributed across banks (lower CVaR relative to VaR) than non-crisis stress, consistent with the HTM channel operating through a systematic factor (interest rate risk) rather than idiosyncratic bank risk. This has direct supervisory implications: system-level stress tests that average across banks may underestimate the true vulnerability, because the HTM channel’s concentration in a few large banks means that system-level CVaR is dominated by tail institutions.

10.5 Uncertainty and Signal Divergence

The uncertainty channel introduces a new dimension to stress testing: not just the magnitude of the shock, but the certainty with which it can be estimated. When LM% word-count sentiment and LLM semantic understanding disagree—what we term “stance distance”—the resulting uncertainty is itself a transmission channel. Stance distance significantly predicts worse bank returns (coef = -0.0091 , $p = 0.005$) and higher cross-sectional dispersion ($p = 0.058$). The ZLB amplification effect (coef = $+0.000332$, $p = 0.051$) suggests that uncertainty is most dangerous when monetary policy is already constrained. The 2013 Taper Tantrum—where Bernanke’s testimony created maximum confusion about the Fed’s QE taper timeline—exhibits the highest stance distance in the ZLB subsample (z-score = 4.58), validating the measure’s ability to capture real-world ambiguity.

11 Conclusion

FOMC statement language is a real-time, publicly available signal that has documented predictive content for the cross-section of bank stress-test returns in both the US and Japan. The signal is sharp only in the pre-DFAST era; after 2008, the DFAST/CCAR framework absorbs the information previously conveyed by FOMC language.

We identify four independent channels through which FOMC language transmits to bank equity returns. During ZLB periods, the NIM compression channel (Dovish $\times$ ZLB $\times$ NIM = $+0.68$, $p < 0.001$) and the CRE sensitivity channel (Dovish $\times$ ZLB $\times$ CRE = $+0.25$, $p = 0.013$) both make banks benefit from dovish FOMC. During rapid rate hikes, the HTM unrealized loss channel (FastHike $\times$ HTM = -0.09 , $p < 0.001$) makes high-HTM

banks suffer disproportionately. The HTM channel dominates the CRE channel, providing a new interpretation of the 2023 SVB failure as an “HTM unrealized loss crisis” rather than a “CRE crisis.”

Three additional analyses strengthen these findings. First, a within-bank quasi-experiment isolates the causal effect of accounting classification: the same bank holds both HTM and AFS securities with identical cash flows, differing only in accounting treatment. The Δ test yields -0.054 ($t = -6.93$), confirming that non-mark-to-market accounting is the causal mechanism. A placebo test using random non-FastHike meetings yields an insignificant Δ , confirming specificity to the rapid rate hike period. Second, the NIM channel’s microfoundation is validated through the rebalancing channel: the four-way interaction $\text{Dovish} \times \text{ZLB} \times \text{NIM} \times \text{TradingRatio} = +0.0048$ ($t = 2.79$) confirms that trading income compensates for NIM compression during ZLB periods. Third, a Kuttner surprise robustness check decomposes the transmission into two complementary mechanisms: the *interest rate mechanism* (HTM and NIM channels are driven by rate surprises) and the *information mechanism* (LM% captures forward guidance and risk assessment beyond the rate decision).

The cross-sectional-to-aggregate mapping translates these bank-level effects into system-level risk measures. Under the FastHike scenario, the system-level cumulative CAR is -0.88% , with the top three banks contributing 47% of the impact. The SVB counterfactual analysis reveals that SVB’s extreme HTM ratio explains 76% of its predicted loss, providing a quantitative validation of the HTM channel’s macro significance.

These findings have five policy implications. First, stress test scenarios should include a “ZLB-to-FastHike transition” scenario that explicitly models HTM unrealized losses. Second, the HTM vs AFS asymmetry suggests that accounting transparency—specifically, requiring mark-to-market disclosure of HTM portfolios—would reduce the information asymmetry that amplifies bank stress during rate hikes. Third, the NIM channel’s strength in regional banks suggests that supervisory attention should focus on traditional lending-oriented banks during ZLB periods, when they appear to benefit from QE but are accumulating hidden interest rate risk. Fourth, the uncertainty channel suggests that stress tests should incorporate a “signal ambiguity” dimension: when LM% and LLM classifications disagree on the directional meaning of an FOMC statement, the resulting uncertainty amplifies bank stress and cross-sectional dispersion. Capital buffers should include an uncertainty surcharge proportional to stance distance. Fifth, the system-level analysis reveals that stress tests averaging across banks underestimate true vulnerability, because the HTM channel’s concentration in a few large banks means that system-level CVaR is dominated by tail institutions.

References

- Michael D. Bauer and Christopher J. Neely. Monetary policy and the information channel. *Journal of Monetary Economics*, 145:1–22, 2025. doi: 10.1016/j.jmoneco.2024.103421.
- Ben S. Bernanke and Kenneth N. Kuttner. What explains the stock market’s reaction to federal reserve policy? *Journal of Finance*, 60(3):1221–1257, 2005. doi: 10.1111/j.1540-6261.2005.00760.x.
- Alan S. Blinder, Michael Ehrmann, Marcel Fratzscher, Jakob De Haan, and David-Jan Jansen. Central bank communication and monetary policy: A survey of theory and evidence. *Journal of Economic Literature*, 46(4):910–945, 2008. doi: 10.1257/jel.46.4.910.
- Board of Governors of the Federal Reserve System. Dodd-frank act stress test publications. <https://www.federalreserve.gov/supervisionreg/stress-tests-capital-planning.htm>, 2024a.
- Board of Governors of the Federal Reserve System. Federal reserve statistical release H.4.1: Factors affecting reserve balances. <https://www.federalreserve.gov/releases/h41/>, 2024b.
- Claudio Borio, Leonardo Gambacorta, and Boris Hofmann. The financial cycle and macroeconomics: What have we learnt? *Journal of Banking & Finance*, 36(11):2994–3010, 2012. doi: 10.1016/j.jbankfin.2012.07.003.
- Yanzhen Chen, Dechang Xu, and Li Wang. Inner confidence: Large language models know when they are right. Working Paper 34965, NBER, 2025a.
- Yanzhen Chen, Dechang Xu, and Li Wang. Ewa-fed: Entropy-weighted aggregation for federated learning with heterogeneous clients. *arXiv preprint*, 2025b. Under review.
- Anna Cieslak, Andreas Schrimpf, and Fabian Visser. The short-term rate, the term structure, and the monetary policy transmission. *Journal of Financial Economics*, 134(3):625–649, 2019. doi: 10.1016/j.jfineco.2019.07.005.
- Itamar Drechsler, Alexi Savov, and Philipp Schnabl. The deposits channel of monetary policy. *Quarterly Journal of Economics*, 132(4):1819–1876, 2017. doi: 10.1093/qje/qjx019.
- Federal Deposit Insurance Corporation. 2024 FDIC risk review. <https://www.fdic.gov/system/files/2024-06/2024-risk-review.pdf>, 2024.
- Mark Gertler. *Monetary Policy and the Credit Channel*. Federal Reserve Bank of Kansas City, 1994.

- Refet S. Gürkaynak, Brian P. Sack, and Eric T. Swanson. The sensitivity of long-term interest rates to economic news: Evidence and implications for monetary policy. *American Economic Review*, 95(1):425–436, 2005. doi: 10.1257/0002828053828446.
- Stephen Hansen and Michael McMahon. Shocking language: What makes central bank communication more effective? *American Economic Review*, 106(11):3319–3344, 2016. doi: 10.1257/aer.20140875.
- Stephen Hansen, Michael McMahon, and Andrea Prat. Transparency and deliberation within the FOMC: A computational linguistics approach. *Quarterly Journal of Economics*, 133(2):801–870, 2018. doi: 10.1093/qje/qjx045.
- Beverly Hirtle, Anna Kovner, James Vickery, and Meru Bhanot. Assessing financial stability: The Comprehensive Capital Analysis and Review. *Federal Reserve Bank of New York Economic Policy Review*, 26(1):1–28, 2020.
- Priit Jeenas. Firm balance sheet liquidity, monetary policy shocks, and investment dynamics. *Working Paper*, 2019. Stockholm School of Economics.
- Kenneth N. Kuttner. Monetary policy surprises and interest rates: Evidence from the Fed Funds futures market. *Journal of Monetary Economics*, 47(3):523–544, 2001. doi: 10.1016/S0304-3932(01)00055-1.
- Yanzhen Li, Hao Wang, and Wei Zhang. Large language models for financial text analysis. *Journal of Financial Economics*, 2025. Forthcoming.
- Tim Loughran and Bill McDonald. When is a liability not a liability? textual analysis, dictionaries, and 10-Ks. *Journal of Finance*, 66(1):35–65, 2011. doi: 10.1111/j.1540-6261.2010.01625.x.
- Yihao Lu and Jing Cynthia Wu. The FOMC and the SEC: How monetary policy shapes bank regulatory capital. *Working Paper*, 2024. University of Notre Dame.
- Hélène Rey. Dilemma not trilemma: The global financial cycle and monetary policy independence. *Federal Reserve Bank of Kansas City Economic Symposium Proceedings*, pages 285–333, 2013.
- Carlo Rosa. The financial market effect of FOMC minutes. *Economic Policy Review*, 19(2):67–87, 2013.
- Yixin Wang, Haotian Liu, and Wei Zhang. Central bank communication in the era of large language models. *Working Paper*, 2025. Forthcoming.